

Coloured complexes of transition element (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- Apply a colour wheel to deduce the wavelengths and frequencies of light absorbed and/or observed.
- Explain why transition metal complexes appear coloured.

Similar to coloured glass like you saw in [section S3.1.8](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-2-id-43595/>), many gemstones appear coloured due to the presence of

transition metal ions in the crystal structure. Transition metal ions interact with visible light in a unique manner causing the appearance of a specific colour for different gemstones. Gemstones that are more vivid in colour have higher value. This means the presence of transition metal impurities increase the value of the stone.

Interactive 1 shows some examples of gemstones, the chemical makeup of the gemstone and the transition metal ion responsible for its colour.

Interactive 1. Gemstones, Their Chemical Makeup and the Transition Metal Ions Responsible for Their Colour.

 More information for interactive 1

An interactive slideshow with seven slides presents photographs of gemstones, along with their chemical formulas and descriptions explaining the transition metal ions responsible for their colors.

Slide 1: Peridot

A cushion-shaped green gemstone.

Chemical formula: Mg_2SiO_4

Description: Color from Fe^{2+} ions replacing some magnesium ions in the crystal structure.

Slide 2: Garnet

A round-shaped red gemstone.

Chemical formula: $\text{Mg}_3\text{Al}_2(\text{SiO}_4)_3$

Description: Color from Fe^{2+} ions replacing some magnesium ions in the crystal structure.

Slide 3: Tourmaline

A rectangular pink gemstone.

Chemical formula: $\text{Na}_3\text{Li}_3\text{Al}_6(\text{BO}_3)_3(\text{SiO}_3)_6\text{F}_4$

Description: Color from manganese ions replacing some lithium and aluminum ions in the crystal structure.

Slide 4: Aquamarine

Trillion-shaped blue gemstones.

Chemical formula: $\text{Be}_3\text{Al}_2(\text{SiO}_3)_6$

Description: Color from Fe^{2+} and Fe^{3+} ions replacing some aluminum ions in the crystal.

Slide 5: Emerald

A deep, vivid green rectangular gemstone with step facets.

Chemical formula: $\text{Be}_3\text{Al}_2(\text{SiO}_3)_6$

Description: Color from manganese ions replacing some aluminum ions in the crystal structure.

Slide 6: Ruby

A pinkish-red, oval-shaped gemstone.

Chemical formula: Al_2O_3

Description: Color from chromium ions replacing some aluminum ions in the crystal structure.

Slide 7: Sapphire

A deep blue, round-shaped gemstone.

Chemical formula: Al_2O_3

Description: Color from titanium and iron ions replacing some aluminum ions in the crystal structure.

This interactivity helps users to understand the chemical formulas of common gemstones and explains how transition metal ions contribute to their characteristic color.

Similar to coloured gemstones, coloured solutions appear the colour they are because of the way they interact with visible light. We learned earlier in [section S3.1.8](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-2-id-43595/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-2-id-43595/>) that transition metal complexes appear coloured. But why is it that they are coloured but others are not? What is unique about the way that transition metal complexes interact with visible light? What ways can we quantitatively evaluate the colour of transition metal complexes?

Recall from [section S3.1.8](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-2-id-43595/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-2-id-43595/>) that complex ions are formed when ligands form coordinate bonds to the transition metal ion through the donation of a lone pair of electrons. As the ligand approaches the transition metal ion, the

electrons from the ligand will be closer to the electrons in some d-orbitals and further away from others. Let's look, in **Figure 1**, at the shapes of the d-orbitals and the shape of an octahedral complex to predict which d-orbitals the ligands would be closer to and which ones it would be further away from.

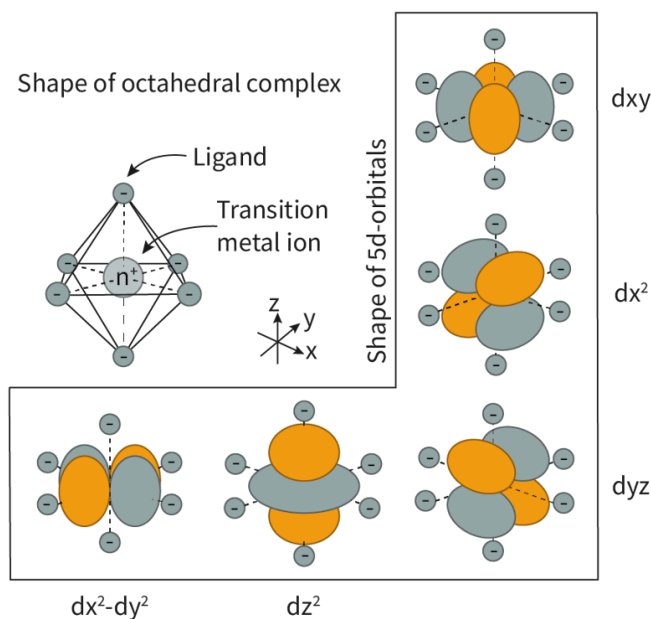


Figure 1. The shapes of an octahedral complex and the five d-orbitals.

 More information for figure 1

The image shows a diagram illustrating the shape of an octahedral complex and the five 3d orbitals: d_{xy} , d_{x^2} , d_{yz} , $d_{x^2-dy^2}$, and d_{z^2} . The octahedral complex is depicted in the top left, showing a central transition metal ion surrounded by six ligands, positioned along the x, y, and z axes.

Adjacent to this, the diagram portrays the shapes of the five 3d orbitals: - **D_{xy} Orbital:** Positioned towards the top right, with lobes oriented between the x and y axes. - **D_{x^2} Orbital:** Below the d_{xy} , with lobes lying directly along the x-axis. - **D_{yz} Orbital:** Further down, oriented between the y and z axes. - **$D_{x^2-Dy^2}$ Orbital:** Located at the bottom-left, with lobes stretching along the x and y axes. - **D_{z^2} Orbital:** To the right of the $d_{x^2-dy^2}$, with lobes elongated along the z-axis.

The d-orbitals are shown with grey and orange lobes indicating the regions of electron density. Some electrons are depicted as grey spheres positioned around the orbitals, emphasizing electron density regions. The orientation of the orbitals corresponds to how they're arranged within an octahedral complex, impacting electron repulsion in complex ion formation.

[Generated by AI]

You can see in **Figure 1** that two of the d-orbitals have electron density in the same region of space as the ligands need to occupy for the formation of a complex ion. This is the $d_{x^2-dy^2}$ and d_{z^2} -orbitals as the electron density within these orbitals aligns along the x, y and z axes as does the arrangement of ligands in an octahedral complex. As these ligands approach a transition metal ion, the electrons in the d-orbitals repel the lone pair electrons on the incoming ligand. This causes the d-

sublevel to split in energy where the orbitals that are closer in proximity to the ligands become higher in energy due to this repulsion and the remaining orbitals become slightly lower in energy. This is illustrated in **Figure 2**.

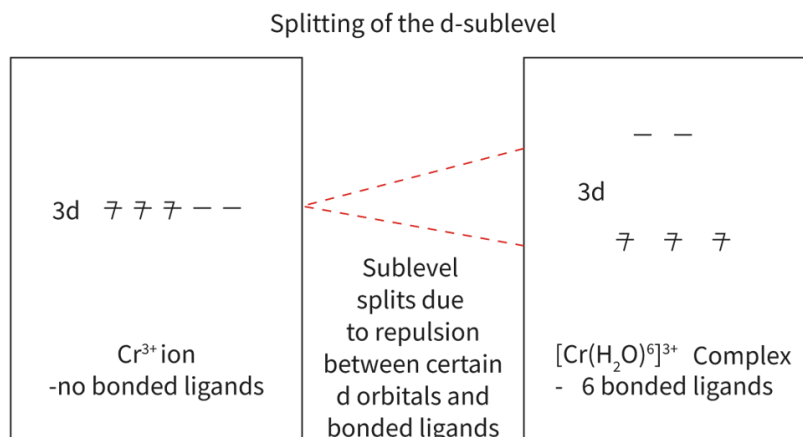


Figure 2. Splitting of the d-sublevel.

 More information for figure 2

The diagram illustrates the splitting of the d-sublevel energy levels in a chromium ion (Cr^{3+}) due to the presence of ligands. On the left, it shows an isolated chromium ion with 3d orbitals at the same energy level, labeled as " Cr^{3+} ion - no bonded ligands." The 3d orbitals are represented by horizontal lines, three of which are at the same energy level.

On the right, the diagram shows how these energy levels split when the ion is part of a complex, labeled as " $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ Complex - 6 bonded ligands." In this part, there are two levels of energy: one with two orbitals higher in energy and the other with three orbitals lower in energy. The splitting is indicated by red dashed lines pointing from the single energy level in the free ion to the two levels in the complex.

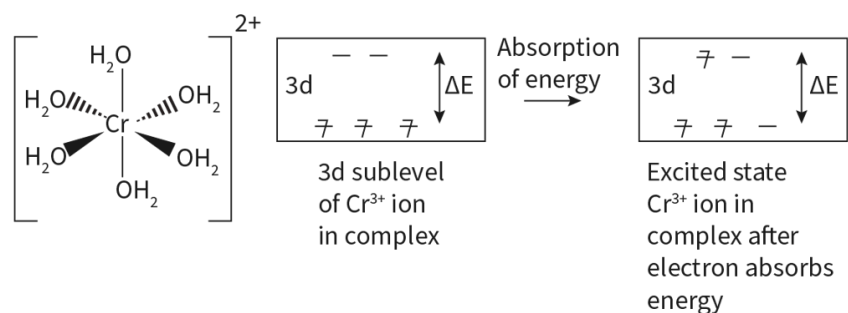
The text notes that the splitting occurs due to repulsion between certain d orbitals and bonded ligands.

[Generated by AI]

Study skills

You do need to know that the d-sublevel splits after the formation of a complex, but you are not required to know which d-orbitals are involved in splitting.

Since the d-sublevel splits into separate energies, this means electrons occupying d-orbitals in the lower level are able to absorb energy and transition to the higher level, forming an excited state. The formation of this excited state transition metal ion is shown in **Figure 3**.

Formation of an excited state Cr^{3+} ion in complex**Figure 3.** Formation of an excited state Cr^{3+} ion in complex.

More information for figure 3

The diagram illustrates the formation of an excited state Cr^{3+} ion in a complex through the absorption of energy. On the left, it shows Cr^{3+} ion surrounded by six water molecules, representing a coordination complex. The ion is under an oxidation state of 2+. Next to it, there are two energy boxes representing the 3d sublevel energy states. The first box shows the ground state, with three pairs of electron spins in lower energy, and an energy gap indicated by ΔE . An arrow labeled "Absorption of energy" points from this ground state box to the right box, which depicts the excited state. In this excited state, one of the pairs of electron spins has transitioned to a higher energy level, leaving two pairs in the lower level, indicating the movement of electrons after absorbing energy.

[Generated by AI]

The difference in energy of the splitting of the d-sublevel in transition metal ions correlates to the energy from photons of visible light. This is illustrated in **Figure 4**. The difference in energy between the split d-sublevels will differ depending on what transition metal ion makes up the complex, what oxidation state the transition element has, the ligands bonded to the ion, and the arrangement of the ligands around the ion.

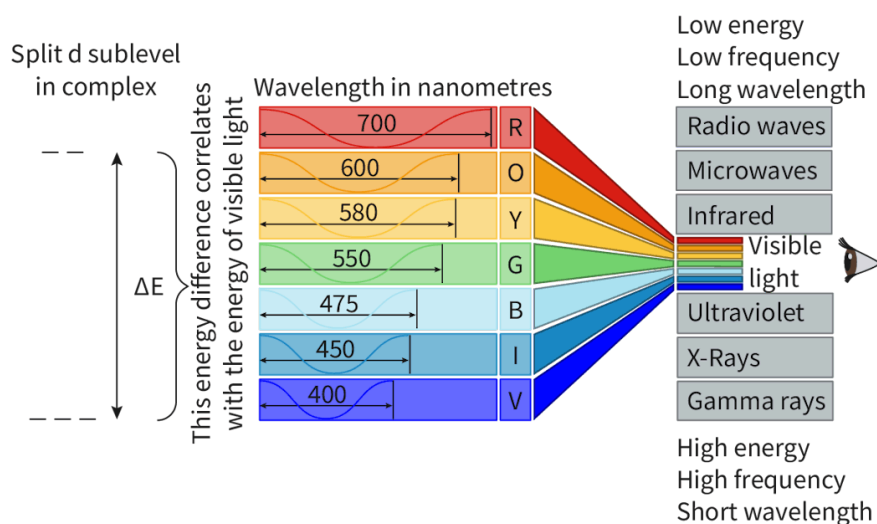


Figure 4. Energy difference between split d-sublevel correlates with visible light.

 More information for figure 4

The image is a diagram that illustrates the correlation between the energy difference in the split d-sublevel of transition metal complexes and the energy of visible light. On the left side, there's a vertical label "Split d sublevel in complex" with arrows indicating an increase in energy difference (ΔE). Beside this is a column labeled "Wavelength in nanometres" with color-coded sections indicating parts of the visible light spectrum: 700 nm (Red), 600 nm (Orange), 580 nm (Yellow), 550 nm (Green), 475 nm (Blue), 450 nm (Indigo), and 400 nm (Violet). Each color band shows double-headed horizontal arrows and sinusoidal wave patterns. On the right, there is a horizontal spectrum showing wavelengths ranging from low energy, low frequency, and long wavelength (Radio waves, Microwaves, Infrared) to high energy, high frequency, and short wavelength (Ultraviolet, X-Rays, Gamma rays). The visible light section is shown in the center of this spectrum.

[Generated by AI]

Since this d-sublevel splitting has the energy difference correlating with visible light, this means that when a solution containing a transition metal complex encounters visible light photons, electrons in the lower d-sublevel of the transition metal ion can absorb specific wavelengths of visible light to form excited state ions. Different complexes will have slightly different energies between the split sublevel resulting in visible light of a different wavelength absorbed for each complex. As specific wavelengths of visible light are absorbed, the remainder wavelengths of visible light are transmitted through the solution. This is illustrated in **Figure 5**.

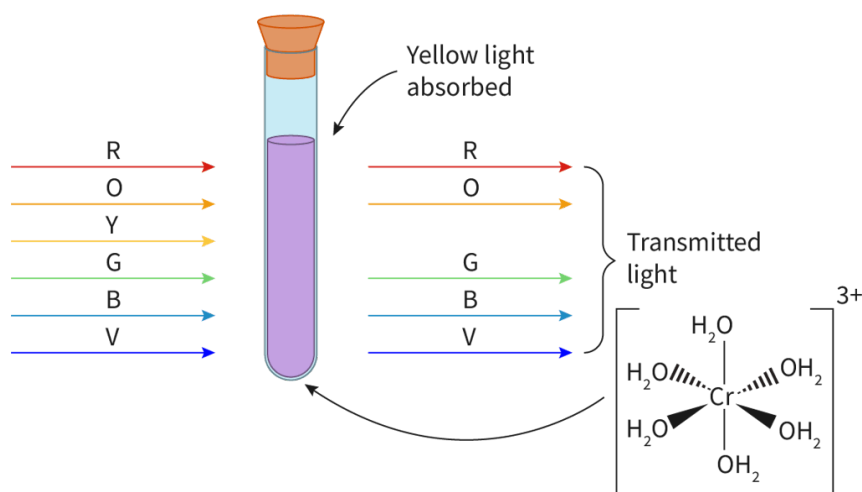


Figure 5. A solution of transition metal complexes absorbing specific wavelengths of visible light.

 More information for figure 5

The image is a diagram illustrating how a solution of transition metal complexes, specifically a chromium complex, absorbs specific wavelengths of visible light. The test tube contains a violet solution, indicating that yellow light (Y) has been absorbed. Arrows labeled as different colors of light, represented as R (Red), O (Orange), Y (Yellow), G (Green), B (Blue), and V (Violet), are shown entering the solution. The yellow light is absorbed, and the remaining

colors are transmitted as indicated by arrows exiting the test tube. The transmitted light refers to the visible light that the eye perceives, shown as an arc labeling the light as 'Transmitted light'. To the right, the chemical structure of the chromium complex $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ is displayed. This diagram helps explain the concept of color absorption and transmission related to transition metal complexes.

[Generated by AI]

The transmitted light is what our eyes detect. If you notice in **Figure 5** yellow light was absorbed even though the solution colour appears violet. So how does the colour and wavelength of a transition metal complex relate to the colour and wavelength of light it absorbs? When we remove wavelengths of yellow light from the visible spectrum, our eye perceives the transmitted light as the complementary colour to yellow, which is violet. We can use the colour wheel to predict the wavelengths and colour a solution absorbs based on the colour a solution appears. A version of the colour wheel can be found in [section 15](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/colour-wheel-with-wavelengths-of-the-visible-id-45117/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/colour-wheel-with-wavelengths-of-the-visible-id-45117/>) of the DP chemistry data booklet and in **Figure 6**. The complementary colour and wavelengths can be found across the colour wheel.

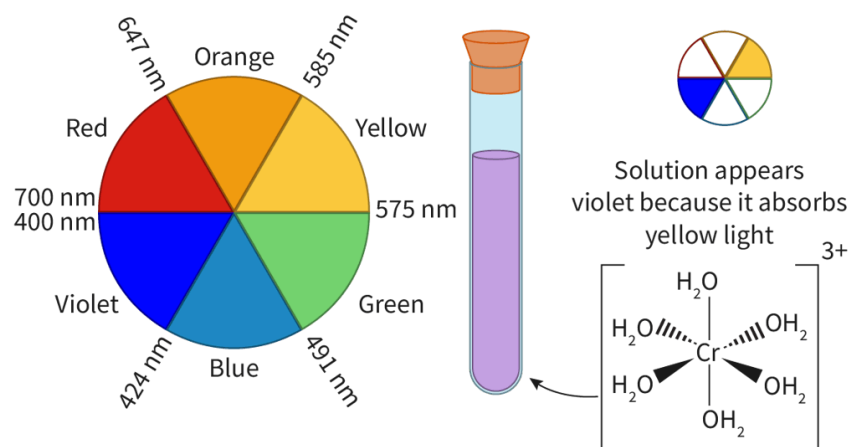


Figure 6. Using the colour wheel to correlate colours of visible light absorbed and perceived by the eye.

 More information for figure 6

The image features a color wheel that is divided into six colored sections around a central point. The sections represent different colors of visible light: red, orange, yellow, green, blue, and violet. Each section is labeled with its corresponding wavelength in nanometers (nm). Specifically, red is 647 nm, orange is unlabeled, yellow is 585 nm, green is 575 nm, blue is 491 nm, and violet is 424 nm. The range from 700 nm to 400 nm is also noted.

Beside the color wheel, there is a test tube containing a violet-colored solution. The image explains visually why the solution appears violet: the solution absorbs yellow light, which is represented on the color wheel opposite violet. An additional small color wheel is shown to illustrate complementary colors.

There is also a molecular diagram representing a chromium complex with six water molecules $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ next to the test tube. The text below it states, "Solution appears violet because it absorbs yellow light."

[Generated by AI]

Study skills

It is important to note that transition metal complexes appear coloured because they absorb visible light. The colour they appear to the human eye is dependent on the wavelengths of light that are transmitted. A common misconception is that transition metal complexes emit light. Excited electrons in transition metal complexes tend to release energy in the form of heat when they return to the ground state rather than light. Ensure on assessments to discuss the colour of transition metal complexes in the context of light absorption and transmission, not emission.

We can relate the energy of splitting to the frequency and wavelength of the light absorbed using two simple equations, both found in [section 1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/some-relevant-equations-id-45103/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/some-relevant-equations-id-45103/>) of the DP chemistry data booklet. You learned how to use these equations in [section S1.3.6](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculating-ionisation-energies-hl-id-44044/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculating-ionisation-energies-hl-id-44044/>). A summary of these equations and the variables used within these equations is given in **Table 1**.

Table 1. Equations relating the difference in energy of orbitals to the frequency and wavelength of light absorbed.

Equation	Symbols	Units
$E = hf$	E = energy	joules (J)
	h = Planck's constant (6.63×10^{-34} J s)	joule seconds (J s)
	f = frequency	per second (s^{-1})
$c = f\lambda$	c = speed of light (3.00×10^8 m s $^{-1}$)	metres per second (m s $^{-1}$)
	λ = wavelength	metres (m)
	f = frequency	per second (s^{-1})

Looking at these equations you can see that the greater the energy difference in the splitting of the d-sublevel the higher frequency light that will be required to be absorbed since energy and frequency are directly proportional. In the second equation we can see that frequency and wavelength are inversely proportional, meaning that the higher frequency light absorbed will be from photos of light with smaller wavelengths. These equations allow us to calculate energy and frequency

from wavelength absorbed and vice versa. The worked example shows the calculation of frequency of light and energy difference of splitting from a solution absorbing yellow light with a wavelength of approximately $\lambda = 580 \text{ nm}$.

Worked example 1

A transition metal complex appears purple and absorbed yellow light with a wavelength of approximately $\lambda = 580 \text{ nm}$.

Calculate the following, expressing your answers to 3 significant figures:

1. The frequency of light absorbed
2. The energy of the d-sublevel splitting

1.

Solution steps

Step 1: Write out the values given in the question.

Step 2: Change the units where necessary.

Step 3: Write out the equation.

Step 4: Rearrange the equation to make f the subject.

Step 5: Substitute the values given.

Step 6: State the answer with appropriate units and the number of significant figures used in rounding.

2.

Solution steps

Ca

Step 1: Write out the values given in the question.

$h =$

$f =$

Step 2: Write out the equation.

E

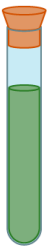
Step 3: Substitute the values given.

E

Solution steps	Ca
Step 4: State the answer with appropriate units and the number of significant figures used in rounding.	E

Try **Interactive 2** to match the d-sublevel splitting and the wavelength of absorption to the test tubes containing solutions of transition metal complexes.

d-sublevel splitting



Complex

Absorbs

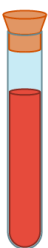
d-sublevel splitting



Complex

Absorbs

d-sublevel splitting



Complex

Absorbs

$\lambda = 450 \text{ nm}$

$\updownarrow \Delta E$

$\lambda = 520 \text{ nm}$

$\updownarrow \Delta E$

$\lambda = 650 \text{ nm}$

$\updownarrow \Delta E$

☒ Check

Interactive 2. Match the d-sublevel splitting and wavelength of absorption to the complex ions.

 More information for interactive 2

An interactive illustration features three test tubes containing coloured complex solutions: green, yellow, and red. Each test tube is closed with a rubber cork. Users need to drag and drop the correct wavelength of absorption and d-sublevel splitting into the correct test tube.

The drag-and-drop options are provided as follows:

Wavelength (λ):

$\lambda = 450 \text{ nm}$

$\lambda = 520 \text{ nm}$

$\lambda = 650 \text{ nm}$

d-orbital splitting diagrams:

Each diagram shows three lower energy orbitals at the bottom and two higher energy orbitals at the top. The vertical distance between them represents ΔE .

Diagram with maximum ΔE

Diagram with medium ΔE

Diagram with minimum ΔE

This interactivity helps users understand the colour of the transition metal complex with the wavelength of light it absorbs and the degree of d-orbital splitting.

Read below for the solution:

Green coloured solution:

Diagram with minimum ΔE

$\lambda = 650 \text{ nm}$

Yellow coloured solution:

Diagram with maximum ΔE

$\lambda = 450 \text{ nm}$

Red coloured solution:

Diagram with medium ΔE

$\lambda = 520 \text{ nm}$

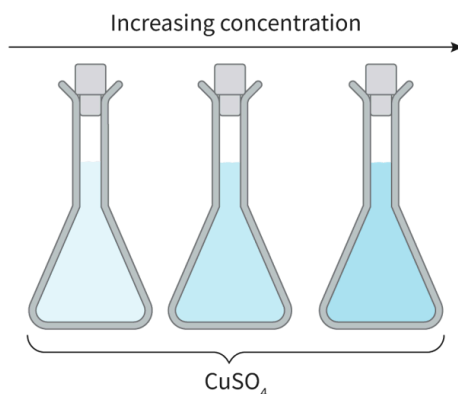
Theory of Knowledge

The colour wheel is an integral tool in the field of art and design to combine colours that are pleasing to the eye. There is evidence of artwork using paint made from ashes and plant materials going back 40 000 years. More recently, paints using lead, cadmium and chromium have been used to provide more vivid colours than natural pigments can provide. It is Sir Isaac Newton, a mathematician, who is credited with arranging colours into a wheel after his experiments with a prism in 1666. Does knowledge of experimental work with prisms validate concepts that artists instinctively understood thousands of years ago?

Quantifying transition metal ions in solution

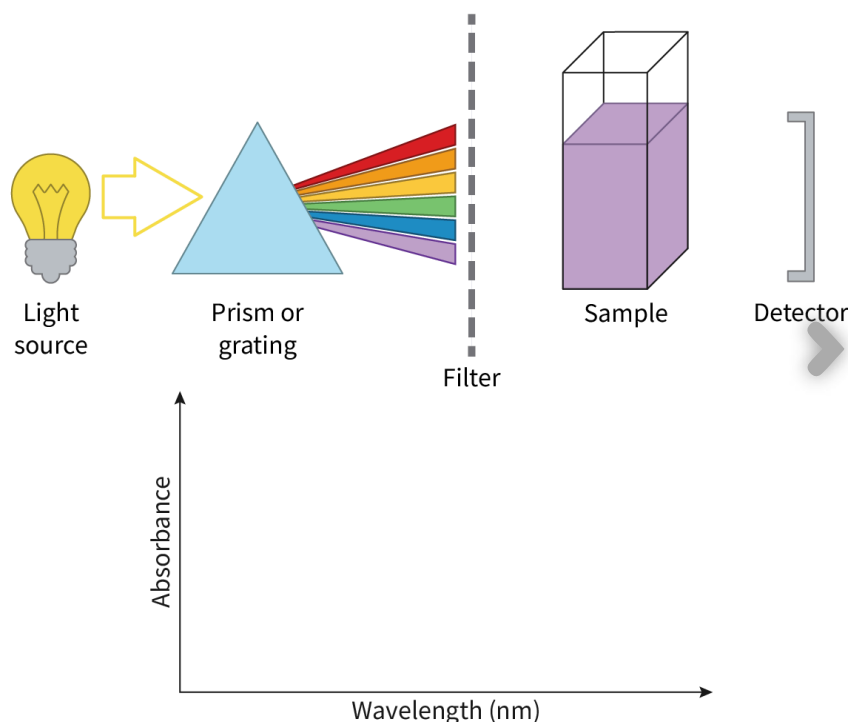
Since transition metal ions formed coloured complexes, we can utilise the colour of the solution to quantify the concentration of transition metal ions in solution. What do you think happens to the appearance of a complex when the concentration is high? What about when it is lower?

Two solutions of $\text{CuSO}_4(\text{aq})$ shown, note what happens to the appearance of the solution colour when the concentration of Cu^{2+} ions increases or decreases.



Two methods that can be used to quantify the concentration of transition metal ions in solution based on the colour they appear are spectrophotometry and colorimetry. **Interactive 3** shows generally how a spectrophotometer works to collect a spectrum of light absorbed.

General set up of spectrophotometer



Interactive 3. How a Spectrophotometer Works to Collect a Spectrum of a Coloured Solution.

 More information for interactive 3

An interactive slideshow presents the general setup of a spectrophotometer and its operation with different wavelengths of light.

Slide 1: Setup

Above: A schematic diagram of a spectrophotometer setup includes a light source, a prism or grating, a filter that allows a specific wavelength of light to pass, a cuvette containing the sample, and a detector that captures the transmitted light.

Below: A graph plots absorbance on the y-axis and wavelength in nm on the x-axis.

The following slides exhibit the graph changes for different colors.

Slide 2: Purple light passing through

Using the prism or grater and the filter, only purple light passes through the sample and reaches the detector.

The graph exhibits a small horizontal bar lying on the x-axis, indicating that no

violet wavelengths are absorbed.

Slide 3: Blue light passing through

Using the prism or grater and the filter, only blue light passes through the sample and reaches the detector.

The graph exhibits a medium-sized horizontal bar lying on the x-axis, indicating that no blue wavelengths are absorbed.

Slide 4: Green light passing through

Using the prism or grater and the filter, only green light passes through the sample and reaches the detector.

The graph exhibits a long horizontal bar on the x-axis, indicating that no green wavelengths are absorbed.

Slide 5: Yellow light passing through

Using the prism or grater and the filter, only yellow light passes through the sample and reaches the detector.

The graph exhibits a long horizontal bar on the x-axis, followed by a concave downward curve. The peak at $\lambda = 580 \text{ nm}$ on the curve indicates that yellow light is absorbed.

Slide 6: Orange light passing through

Using the prism or grater and the filter, only orange light passes through the sample and reaches the detector.

The graph exhibits a long horizontal bar lying on the x-axis, followed by a concave downward curve and a small horizontal bar. The small horizontal bar after the peak indicates that no orange light is absorbed.

Slide 7: Red light passing through

Using the prism or grater and the filter, only red light passes through the sample and reaches the detector.

The graph exhibits a long horizontal bar lying on the x-axis, followed by a concave downward curve and a small horizontal bar. The small horizontal bar after the peak indicates that no red light is absorbed.

This slideshow helps users understand the spectrometer setup and observe light absorption trends.

Both the spectrophotometer and colorimeter provide the absorbance of light at specific wavelengths. A spectrophotometer analyses all the wavelengths in the visible spectrum while a colorimeter only analyses specific wavelengths unique to the physical instrument. **Figure 7** shows how a spectrum produced by a spectrophotometer would change as the concentration of the solution changes.

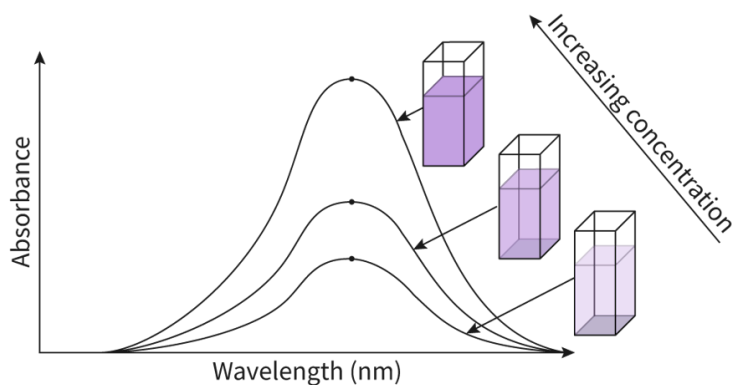


Figure 7. How a spectrophotometry spectrum changes with concentration.

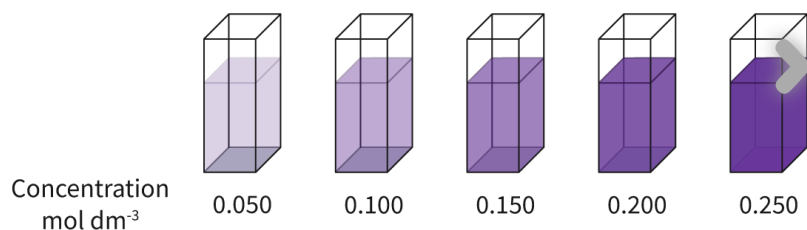
 More information for figure 7

The graph displays the relationship between absorbance and wavelength. The X-axis is labeled 'Wavelength (nm)' and the Y-axis is labeled 'Absorbance.' There are multiple curves representing different concentrations. Each curve shows that absorbance increases with wavelength up to a peak, and then decreases. The peaks of these curves are progressively higher with increasing concentration, which is visually represented by three purple boxes positioned above the curves. An arrow labeled 'Increasing concentration' points diagonally across the graph, illustrating that absorbance increases with concentration.

[Generated by AI]

But how exactly does knowing the absorbance of light help us quantify the concentration of transition metals in solution? This is where we need to utilise some known information for comparison. If we know the identity of the sample, we can make up solutions of known concentrations. We can analyse the absorbance of light for each of the concentrations and compare the absorbance of the unknown concentration to the known concentrations. This enables us to deduce the concentration. **Interactive 4** shows the process for how to do this.

Use spectrophotometer or colorimeter to deduce the absorbance at specified wavelength for each sample



Interactive 4. Deducing the Unknown Concentration of a Sample Using Spectrophotometry.

 More information for interactive 4

An interactive slideshow with three slides demonstrates the process of deducing the unknown concentration of a sample using spectrophotometry. Each slide contains explanatory text along with relevant visuals to illustrate key concepts.

Slide 1:

The text reads as follows: Use a spectrophotometer or colorimeter to deduce the absorbance at a specific wavelength for each sample.

Below are illustrations of five cuvettes containing a purple solution at different

concentrations.

The concentrations of each cuvette, in mol dm^{-3} , are as follows:

Cuvette 1- 0.050

Cuvette 2- 0.100

Cuvette 3- 0.150

Cuvette 4- 0.200

Cuvette 5- 0.250

Slide 2:

The text reads as follows: Plot the absorbance versus concentration for each sample and add the line of best fit.

A scatterplot displays absorbance at 580 nm on the y-axis and concentration on the x-axis. Five data points are marked in an increasing trend, with a best-fit line passing through them. This plot is referred to as the calibration curve.

Slide 3:

The text reads as follows: Use a spectrophotometer to analyze absorbance at the same wavelength of the sample with unknown concentration. Use the line of best fit to determine the concentration for the absorbance value.

An illustration displays a cuvette containing a solution with an unknown concentration on the left. A scatterplot on the right displays absorbance at 580 nm on the y-axis and concentration on the x-axis. Five data points are marked in an increasing trend, with a best-fit line passing through them. A red dashed line extends from the absorbance value on the y-axis to the best-fit line, then continues downward to mark the corresponding unknown concentration on the x-axis.

This slideshow helps users understand how spectrophotometry is used to determine the concentration of an unknown solution.

Practical skills

Tool 1: Experimental techniques — Applying techniques

Analysing complexes using spectrophotometry or colorimetry is an excellent way to study colourful solutions in a data rich manner. We can use these techniques to determine the concentration of solutions containing transition metal ions, to study the wavelength of maximum absorbance as binding ligands are exchanged, or even to test methods to remove metal ions from water sources. Read through the abstract of the following [journal article](https://pubs.acs.org/doi/10.1021/acssuschemeng.5b00207)  (<https://pubs.acs.org/doi/10.1021/acssuschemeng.5b00207>). Can you think of how you might use spectrophotometry in this investigation?

The following activity has some questions for you to consider relating to transition metal complexes.

Activity

- **IB learner profile attribute:** Knowledgeable

- **Approaches to learning:** Communication skills — Presenting data appropriately
- **Time required to complete activity:** 30—45 minutes
- **Activity type:** Individual activity

The following activity has some questions for you to consider. You will need to write out your answers to the questions asked along with accessing a simulation and graphing data.

Part 1: Coloured solutions

Figure 8 shows two beakers. One containing ZnSO_4 the other containing CuSO_4 . Explain, using electron configurations and d-sublevel splitting, why solutions containing the Cu^{2+} ion is coloured while the one containing Zn^{2+} ions is not coloured.

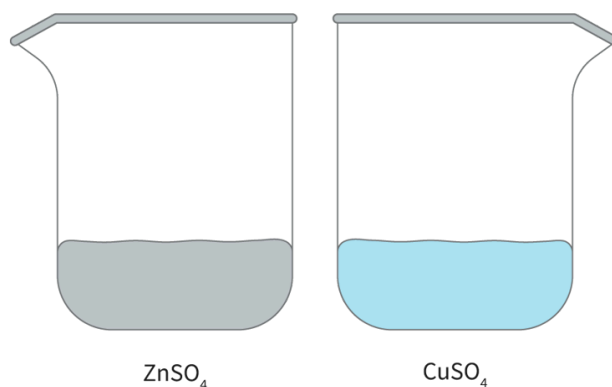


Figure 8. Beakers containing ZnSO_4 and CuSO_4 .

Part 2: Spectrophotometry to deduce unknown concentration

A sample of water, believed to be contaminated with cobalt(II) chloride was found. You can find the concentration of cobalt(II) chloride in your unknown sample by first analysing solutions of five different concentrations of cobalt(II) chloride. Using spectrophotometry, you collect the data shown in **Table 2**.

Table 2. Data collected using spectrophot

Concentration (mol dm^{-3})	Abs
0.0550	
0.100	
0.135	
0.170	
0.185	

Using this data complete the following:

- Construct a plot with values for concentration on the x-axis and values for the absorbance at $\lambda = 549 \text{ nm}$ on the y-axis.
- Add a trendline and an equation of the line.
- Solve for the concentration of a solution of CoCl_2 with an absorbance of 1.19 at $\lambda = 549 \text{ nm}$.
- Compare your calculated concentration with a peer.

Part 2: Ligand exchange

By adding different substances, it is common for the ligands bonded to transition metal ions to exchange for different ligands. The ability for ligands to exchange depends on the strength of the interaction between the lone pair of electrons on the ligand and the transition metal ion and the quantity of ligand molecules surrounding transition metal ions (**Table 3**).

Table 3. Ligand exchange data.

Substance	Substance added	Ligand(s)	Wavelength
$\text{CuCl}_2(\text{aq})$	none	Cl^-	
	H_2O	H_2O	
	KSCN	SCN^-	
	NH_3	NH_3	

- Rank the ligands in order of increasing energy between the split d-sublevels of the Cu^{2+} ion. Explain how you used the data in **Table 3** to do so.
- Calculate the frequency (in s^{-1}) of light absorbed for each Cu^{2+} complex.
- Calculate the energy (in J) of the splitting between the d-sublevels for each Cu^{2+} complex.